Vehicular Communications 52 (2025) 100887

Contents lists available at ScienceDirect

Vehicular Communications

journal homepage: www.elsevier.com/locate/vehcom

Detection of zero-day attacks via sample augmentation for the Internet of Vehicles

Bingfeng Xu [a], Jincheng Zhao [a], Bo Wang [a], Gaofeng He [b],*

[a] College of Information Science and Technology & Artificial Intelligence, Nanjing Forestry University, Nanjing, 210037, China
[b] School of Internet of Things, Nanjing University of Posts and Telecommunications, Nanjing 210003, China

ARTICLE INFO

Keywords:
Internet of Vehicles
Zero-day attack detection
Sample augmentation
Generative adversarial network
Few-shot learning

ABSTRACT

Detecting zero-day attacks is a critical challenge in the Internet of Vehicles (IoV). Due to the limited availability of labeled attack data, anomaly-based methods are predominantly employed. However, the variability in the driving environment and behavioral patterns of vehicles introduces significant fluctuations in normal behavior, which in turn leads to high false positive rates when using these methods. In this work, we propose a novel detection method for zero-day attacks in IoV through sample augmentation. We first analyze the similarities between known and zero-day attacks in IoV. Based on the analysis, a Few-shot Learning Conditional Generative Adversarial Network (FLCGAN) model with multiple generators and discriminators is developed. Within this framework, an attack sample augmentation algorithm is designed to enhance input data by expanding the known attack dataset, thereby reducing false positives. To address the data imbalance caused by the limited number of input attack samples, an ensemble focal loss function is incorporated into the generator to ensure diversity and dispersion of the generated samples. Additionally, a collaborative focal loss function is introduced into the discriminator to improve the classification of difficult-to-classify data. A theoretical analysis is also conducted on the coverage of samples generated by the model. Extensive experiments conducted on the IoV simulation tool Framework For Misbehavior Detection (F2MD) demonstrate that the proposed method surpasses existing approaches in both detection effect and detection delay for zero-day attacks.

1. Introduction

In the Internet of Vehicles (IoV) environment, vehicles in motion have the capability to communicate not only with other vehicles but also with various entities such as infrastructure, pedestrians, and smart devices [1]. This openness enables IoV to play a pivotal role in intelligent transportation systems. However, it simultaneously leads to an escalation in the occurrence of cyberattacks [2]. A notable instance is the 2022 Hyundai Santa Fe hack, which exploited a zero-day vulnerability in the vehicle's remote monitoring software, enabling unauthorized remote control of the door locks, windows, and fuel system [3]. Similarly, at the 2024 Pwn2Own Automotive competition, researchers discovered dozens of zero-day vulnerabilities within just two days, compromising a range of electric vehicle chargers, operating systems, and Tesla components [4]. According to a study by the Ponemon Institute, 80% of security breaches are attributed to zero-day attacks [5]. Evidently, zero-day attacks have emerged as a critical threat in the field of IoV security [6]. Furthermore, zero-day vulnerabilities are security flaws that have yet to be discovered and remedied by software vendors. Consequently, they are challenging to prevent in advance and extremely destructive, potentially causing traffic accidents and endangering the safety of passengers and pedestrians. Therefore, the detection of zero-day attacks has become an urgent and pressing issue within the domain of IoV.

Due to the scarcity of zero-day attack data, intrusion detection in IoV predominantly relies on anomaly-based methods. These methods utilize known normal data to construct models that encapsulate the features of typical behavior, identifying anomalies by detecting differences between observed samples and the established models [7]. Yang et al. proposed a novel multilayer hybrid intrusion detection system for IoV, which employs an anomaly-based intrusion detection system integrating CLKmeans and a biased classifier to detect zero-day attacks [8]. However, the lack of labeled attack data leads to a high false positive rate with this method. To address this issue, Liu et al. proposed a collaborative trust-aware anomaly detection method that uses limited attack data to define a collaborative trust factor for evaluating information security [9]. Nevertheless, the diversity of operational environments and

* Corresponding author.
E-mail addresses: bingfengxu@njfu.edu.cn (B. Xu), zhaojincheng@njfu.edu.cn (J. Zhao), bowang@njfu.edu.cn (B. Wang), hegaofeng@njupt.edu.cn (G. He).

https://doi.org/10.1016/j.vehcom.2025.100887
Received 22 October 2024; Received in revised form 20 December 2024; Accepted 19 January 2025

Available online 21 January 2025

the variability of behavior patterns within IoV can lead to significant deviations in normal behavior. Consequently, anomaly detection based solely on an anomaly-based approach may incorrectly classify these normal variations as anomalies, resulting in high false positive rates.

In practical IoV environments, the fundamental principles underlying remote attacks that exploit software vulnerabilities are similar between zero-day attacks and known attacks [10]. For example, the FragAttacks vulnerability enables attackers to compromise target devices via Wi-Fi networks, gaining control and executing malicious operations [11]. Attackers can leverage this vulnerability to sequentially perform vulnerability exploitation, intrusion attempts, gain access, take control, and execute operations. This process is similar to the attack chain in other known attacks, where control over the target system is gradually acquired by exploiting software vulnerabilities. Due to these similarities, some research has explored the use of transfer learning in this context. Sameera et al. proposed a manifold alignment method that employs transfer learning for zero-day attack detection, transforming the source and target domains into a common latent space and applying a deep neural network to build a framework for zero-day attack detection [12]. Similarly, Mehedi et al. introduced an intrusion detection method based on transfer learning, which uses labeled attack samples to distinguish between normal and malicious data, and designed a deep transfer learning ResNet model for zero-day attack detection [13]. However, these methods are challenged by data bias, which may cause the model to overly emphasize the features of known attack data, thus requiring a substantial amount of such data to achieve satisfactory detection performance. In IoV environment, the scarcity and difficulty of obtaining labeled known attack data further exacerbate this issue. Therefore, these methods are not well-suited for direct application in IoV domain.

In recent years, conditional generative adversarial networks (CGANs) have emerged as a promising approach for mitigating the data bias problem by enabling the generation of potential zero-day attack data from existing samples. This capability allows the model to better capture the novel features of zero-day attacks [14]. Consequently, this paper introduces a novel zero-day attack detection method for IoV through sample augmentation, leveraging few-shot learning conditional generative adversarial networks, referred to as FLCGAN (Few-shot Learning Conditional Generative Adversarial Networks). The proposed method begins with the development of a CGAN model featuring multiple generators and discriminators. Within this framework, an attack sample data augmentation algorithm is designed to enhance the dataset by iteratively selecting a small number of samples with the highest uncertainty scores, thereby reducing false positive rates. Moreover, to address the data imbalance issue caused by the limited number of input attack samples, an ensemble focal loss function is designed for the generator. This function ensures that the generated samples are statistically consistent with real data and exhibit diversity. For the discriminator, a collaborative focal loss function is introduced to enable multiple discriminators to operate in unison, focusing on the classification of challenging samples. On this basis, a theoretical analysis of the coverage of adversarial samples generated by the model was carried out. Finally, extensive experiments are conducted on the F2MD IoV simulation platform to validate the effectiveness of the proposed method in detecting zero-day attacks in real-world IoV environments. The experimental results demonstrate that the FLCGAN model outperforms existing methods in terms of detection performance and false positive rate, and extensive ablation experiments also verify the effectiveness of each component of the model. The contributions of this paper are as follows.

- In this paper, a few-shot learning conditional generative adversarial network model termed FLCGAN is introduced. The FLCGAN model incorporates an attack sample data augmentation algorithm, which enhances known attack samples to optimize the model's input and thereby reduce false positive rates. Furthermore, to mitigate the data imbalance problem caused by the limited number of input attack samples, an ensemble focal loss function is designed for the generator to ensure diversity and dispersion of the generated samples. Concurrently, a collaborative focal loss function is developed for the discriminator to prioritize the classification of challenging samples.
- This paper investigates the similarities between zero-day attacks and known attacks in terms of their characteristics and patterns, thereby establishing a foundation for employing transfer learning in zero-day attack detection. Building on this theoretical framework of similarity, FLCGAN is analyzed, demonstrating that the proposed method possesses the capability to effectively encompass a broader range of zero-day attacks.
- Extensive experiments are conducted on four IoV zero-day attack datasets to assess the performance of the FLCGAN model. The comparison results indicate that FLCGAN surpasses existing methods in both the effectiveness of zero-day attack detection and detection latency. Furthermore, ablation experiments validate the efficacy of the individual components of the FLCGAN model.

The structure of this paper is as follows: Section 2 reviews the related work, Section 3 analyzes the similarities between zero-day attacks and known attacks and presents the FLCGAN zero-day attack detection method, Section 4 introduces the F2MD simulation platform, datasets, and experimental results, Section 5 discusses the proposed method, and Section 6 offers conclusions and outlines future work.

2. Related work

In this section, we review the related work on zero-day attack detection. We begin by briefly discussing the research on anomaly-based and transfer learning-based zero-day attack detection in Sections 2.1 and 2.2. Subsequently, we highlight recent advancements in zero-day attack detection within IoV.

2.1. Anomaly-based zero-day attack detection

Anomaly-based methods are currently prevalent in the field of zero-day attack detection, encompassing primarily density-based [15], clustering-based [16], distance-based [17], and ensemble detectors [18]. Support Vector Data Description (SVDD) is a traditional density-based method that constructs a hypersphere to represent the distribution of normal data, identifying anomalies by distinguishing them from this hypersphere [15]. Expanding upon this approach, reference [19] introduces Deep-SVDD, an extension of SVDD into its deep form, which emphasizes data compactness and distribution over simple categorical classification. Similarly, iForest [20], a conventional ensemble detector-based attack detection method, utilizes the isolation of samples within a tree structure to identify anomalous attacks. However, in high-dimensional or non-linearly separable data spaces, it frequently suffers from high false positive rates. To address this limitation, Xu et al. proposed the Deep-iForest method, which employs randomly initialized neural networks to map raw data into a set of random representations, subsequently partitioning the data through random axis-parallel cuts [21]. This representation scheme enhances the degrees of freedom in partitioning the raw data space, facilitating nonlinear segmentation across subspaces of varying sizes, thereby fostering a unique synergy between random representations and isolation-based random segmentation.

Additionally, several anomaly-based methods have incorporated deep learning techniques. For instance, Yang et al. introduced a zero-day attack detection method based on autoencoders [22]. This approach constructs an autoencoder trained on normal network traffic data, with the anomaly score determined by calculating the weighted sum of the reconstruction loss and Mahalanobis distance of the samples. Network anomalies are effectively detected by applying a threshold to this score. Building on this concept, Hyoseon et al. developed a multilayer autoencoder model featuring multiple detection stages, each utilizing the

L1 norm and Mahalanobis distance to identify attacks [23]. Similarly, Liu et al. proposed an unsupervised generative adversarial networks with multiple generators for outlier detection [24]. In this model, potential outlier samples are selected from a uniform distribution and used to train the discriminator, effectively transforming it into a binary classifier for outlier detection. However, this model is vulnerable to noise. To mitigate this issue, Zhang et al. introduced the self-training STAD-GAN model, which is optimized through a self-training teacher-student framework. In this framework, the teacher model generates reliable, high-quality pseudo-labels, and the student model is iteratively trained on the enhanced dataset, thereby reducing the impact of noise and progressively improving the anomaly classifier's performance [25]. Nonetheless, anomaly-based detection methods depend heavily on a rigid definition of normal behavior, which often results in high false positive rates when encountering novel attacks.

2.2. Transfer learning-based zero-day attack detection

The principles underlying attacks that exploit software vulnerabilities, including known vulnerabilities and zero-day vulnerabilities, are often similar [10]. Leveraging this similarity, several studies have applied transfer learning techniques for zero-day attack detection [26]. For example, Iman et al. proposed a novel intrusion detection system based on transfer learning, which utilizes an optimized one-dimensional convolutional neural network and long short-term memory network to automatically extract spatial and temporal features from raw data, thereby achieving zero-day attack detection through knowledge transfer [27]. Similarly, Upender et al. proposed a deep transfer learning framework that incorporates domain-unified supervised manifold alignment, successfully enabling effective zero-day attack detection by sharing a small number of newly labeled attack examples within the intrusion detection network [28]. To address the scarcity of labeled target instances, Sameera et al. generated target soft labels through clustering and utilized deep neural networks to develop a zero-day attack detection framework [12].

Additionally, Michau et al. proposed a fully unsupervised method that combines adversarial deep learning to transfer and integrate tasks involving complementary training data, thereby enhancing transfer learning representation capabilities and achieving efficient attack detection [29]. Meanwhile, to facilitate the deployment of intrusion detection systems in heterogeneous IoT (Internet of Things) environments, Mehedi et al. designed a transfer learning-based residual neural network. This model, trained on IoT data generated by diverse IoT sensors, effectively detects attacks and demonstrates strong performance against common IoT zero-day attacks such as DoS, DDoS, and Cross-Site Scripting (XSS) [13]. However, these methods often depend on a substantial number of known attack samples to attain high detection accuracy. In the field of IoV, the availability of attack samples is typically limited, further exacerbating the challenges in this field.

2.3. Zero-day attack detection for IoV

Recently, there have been several research efforts focused on zero-day attack detection in IoV. For example, to enhance the security of external communications for autonomous and semi-autonomous vehicles, Ali et al. proposed an intelligent protection mechanism based on the proportional overlap score method, ensuring the safety of these vehicles' external communications [30]. Building on this, Yang et al. addressed the security of both in-vehicle and external networks by combining unsupervised learning with an improved K-means clustering algorithm. They employed two biased classifiers and a Bayesian-optimized Gaussian process to model the various attack and normal patterns present in the data samples [8]. To ensure timely updates to the attack detection model, Li et al. proposed a cloud-assisted IoV update scheme, where pseudo-labels for unlabeled data from new attacks are generated through pre-classification, and these pseudo-labeled data are used for multiple rounds of transfer learning [31]. Additionally, to safeguard sensitive data privacy in IoV, Boualouache et al. proposed the Zero-X framework, which combines open set recognition with privacy-preserving federated learning [32]. To address the security and privacy attacks faced by connected autonomous vehicles (CAVs) on 5G and advanced networks, the team also proposed a novel detection mechanism utilizing a deep autoencoder, which relies exclusively on benign network traffic patterns to detect attacks [33].

Although prior research has shown some success in detecting zero-day attack within IoV, the inherent complexity of the vehicular environment, coupled with the variability in vehicle behavior models, introduces substantial deviations in normal vehicle patterns. Consequently, these factors contribute to elevated false positive rates in such studies. Furthermore, the heavy reliance on a large number of known attack samples poses a major challenge, particularly in IoV domain, where the availability of attack samples is typically limited, further exacerbating this issue. To address the scarcity of attack samples, conditional generative adversarial networks (CGANs) provide a viable solution for sample augmentation through few-shot learning. He et al. designed a CGAN-LSTM model as a distributed collaborative intrusion detection system, integrating LSTM into CGAN training to enhance detection performance under conditions of sample imbalance [34]. In this paper, we propose a zero-day attack detection method for IoV through sample augmentation, leveraging a few-shot learning CGAN. This approach generates potential zero-day attack data using a limited number of known attack samples and incorporates an ensemble focal loss function in the generator to improve the quality of generated samples. Additionally, a collaborative focal loss function is also designed for the discriminator, enabling it to focus on handling difficult-to-classify data. Experimental results demonstrate that this method enables the model to detect zero-day attacks with only a small number of samples, achieving a high detection rate and low false positive rate.

3. Methodology

In this section, we first analyze the similarities between zero-day attacks and known attacks and propose a framework, FLCGAN, for detecting zero-day attacks in IoV through sample augmentation based on this theoretical foundation. An attack data augmentation algorithm is then designed to enhance the input samples within the framework. Additionally, a collaborative focal loss function is designed for the discriminators, while an ensemble focal loss function is designed for the generators. Finally, a theoretical analysis of the coverage of adversarial samples generated by the model is conducted.

3.1. Similarities between zero-day and known attacks

In the domain of IoV, despite the distinct characteristics and varying security challenges posed by zero-day and known attacks, a significant similarity exists between them. This similarity serves a critical theoretical foundation for leveraging the features of known attacks to detect zero-day attacks. Specifically, both known and zero-day attacks operate by identifying and exploiting vulnerabilities within IoV system to execute their attacks [35].

First, regarding attack behavior patterns, both known and zero-day attacks exhibit similar characteristics. For example, the RandomSpeed attack intercepts and alters communication packets between a vehicle and the server, randomly adjusting the vehicle's speed to deceive the server. Drawing inspiration from the RandomSpeed attack, a new type of zero-day attack, termed RandomPosOffset, has emerged. This attack manipulates the vehicle's location information in communication packets by adding a random offset, thereby covertly compromising the vehicle system's security. Moreover, significant similarities are observed in injection attacks for both known and zero-day attacks. A DoS attack, for example, disrupts normal communication between legitimate devices and the server by injecting malicious data during the delay, thereby

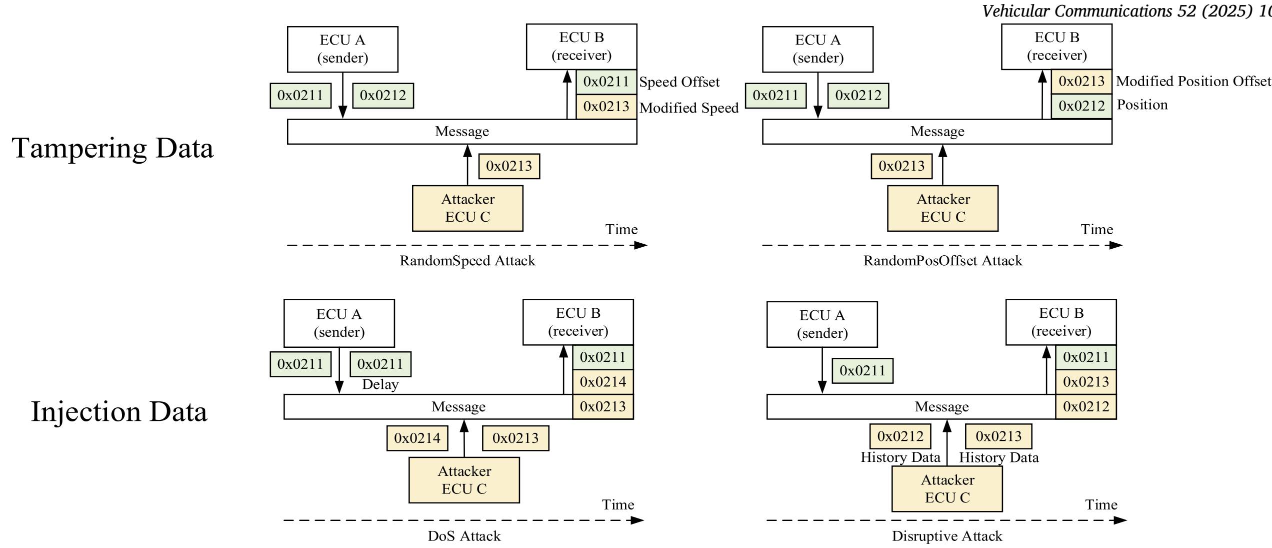

Fig. 1. Similarities Between IoV Attacks.

interfering with the system's normal operation. The disruptive attack, a zero-day variant of the DoS attack, eavesdrops on communication between other devices and the server, injecting outdated beacon data into the communication packets to deceive the system. The similarity in these attack patterns indicates that zero-day and known attacks employ analogous strategies when exploiting network vulnerabilities. In Fig. 1, a schematic diagram of these two types of attacks is presented. As illustrated, despite the distinct characteristics of zero-day and known attacks, there is a notable similarity in their methods of exploiting vulnerabilities to carry out attacks.

Empirical experiments have demonstrated, through Wasserstein distance measurements, that the feature distributions of most zero-day and known attacks exhibit a highly degree of similarity, with Wasserstein distance values below 0.2 [36]. A lower Wasserstein distance value indicates a smaller difference between the two distributions, signifying that zero-day and known attacks overlap substantially in the feature space. This significant overlap highlights their similarity and provides a theoretical foundation for detecting zero-day attacks using features of known attacks.

3.2. Design details

Building on the aforementioned similarities, we can transfer the features of known attacks to the detection of zero-day attacks. By analyzing the behaviors and patterns of known attacks, common features can be identified and applied to enhance the recognition of zero-day attacks. To achieve this, this section introduces a novel zero-day attack detection model, FLCGAN, which is based on multiple generators and discriminators. This model extends the network structure from a single generator and discriminator to multiple ones, enabling it to more effectively capture the complex new features of zero-day attacks and contribute to the stability of GAN training [37]. The structure of the FLCGAN model is depicted in Fig. 2.

Specifically, FLCGAN consists of five primary components: (i) Multiple Generators, which generate adversarial samples that challenge the discriminators' ability to distinguish; (ii) Multiple Discriminators, tasked with evaluating the authenticity of the generated samples; (iii) Attack Sample Data Augmentation, a component that continuously learns from the existing training data distribution, generating potential anomalous points near decision boundaries to enhance the representativeness of the training data; (iv) Ensemble Focal Loss Function, applied to the generators to ensure diversity and dispersion of the generated samples; and (v) Collaborative Focal Loss Function, applied to the discriminators to promote collaboration by assigning higher weights to difficult-to-classify samples, thus mitigating the issue of sample imbalance.

In FLCGAN, each generator receives a random input variable z_i^f along with an embedded vector of class labels y_i^f as input and generates an output $x^f = G(z_i^f, y_i^f)$. The final output is computed as the weighted sum of the outputs from all generators, expressed as $\sum_{j=1}^{N} \pi_j G(z_i^f, y_i^f)$. The discriminator D in FLCGAN consists of two primary components: (i) Adversarial Discriminator: This component is tasked with distinguishing between real and generated data. Real data is selected using the attack sample data augmentation method, while fake data is generated by the generator G. During the discrimination process, conditional information for the labels is incorporated as an embedded conditional vector, and the result of its inner product with the feature vector is used as the discrimination outcome. (ii) Auxiliary Discriminator: This component classifies the discriminator's input as either belonging to an anomaly class or a normal class. In this classification process, the average anomaly score across all the discriminators $\overline{\varphi(x_t)}$ is compared to a predetermined anomaly threshold τ. If the anomaly score exceeds τ, the sample is classified as anomalous; otherwise, it is deemed normal. This multi-generator and multi-discriminator modeling structure significantly enhances the adversarial learning process by generating high-quality samples. Furthermore, when integrated with the focal loss function detailed in the subsequent section, it effectively mitigates the challenge of sample imbalance.

3.2.1. Attack sample data augmentation

Noise can lead the model to make inaccurate judgments, potentially disrupting the training process and impeding its ability to reliably detect attacks. To address this issue, this section proposes an attack sample data augmentation method. This method mitigates noise by filtering out irrelevant data and selecting the most representative attack samples, after which data augmentation is applied to the selected samples, thereby optimizing the input data for FLCGAN. The detailed algorithm is provided in Algorithm 1.

The algorithm improves data quality by selecting attack samples that are closest to the decision boundary while filtering out noise distant from this boundary, thereby reducing false positives. The specific process is as follows: Given a batch of original real data, the average anomaly score of the original data is initially computed using all the auxiliary classifiers in the discriminator. Then, a density-weighted uncertainty sampling method is applied to calculate the uncertainty score for each sample, which incorporates both the sample's density and uncertainty. Samples with the highest uncertainty scores, which are located in high-

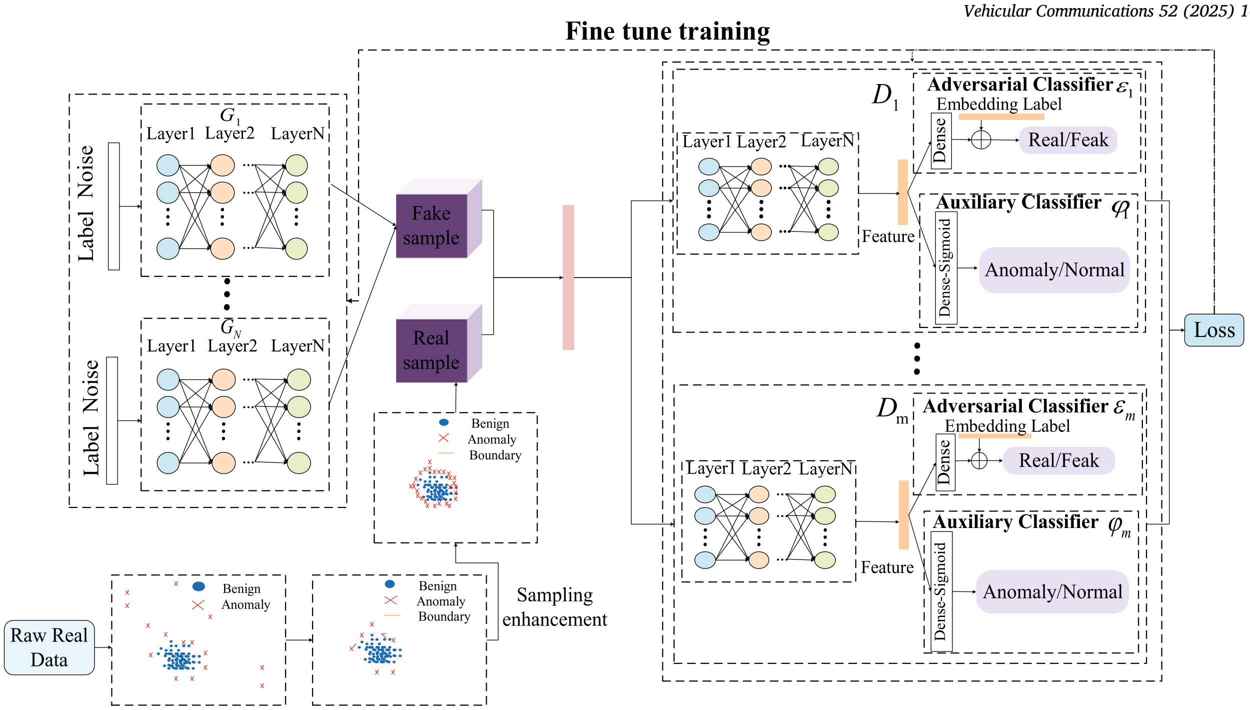

Fig. 2. The network structure of the FLCGAN model: G_i: the i-th generator ($1 \le i \le n$), D_j: the j-th discriminator ($1 \le j \le m$). ε: adversarial classifier used to distinguish between real and fake data, φ: auxiliary classifier used to predict whether input data is an anomaly, $\oplus$: inner product between two vectors.

Algorithm 1: Attack Sample Data Augmentation.

Input: Original n-dimensional data
Output: New synthesized samples E

Evaluate uncertainty scores using density-weighted sampling;
Select top m_s samples with highest scores;
Identify borderline samples using k-nearest neighbors, focusing on majority class neighbors;
Determine imbalance degree $d = \frac{m_s}{m_t}$ and set maximum tolerance threshold d_{th};
if $d < d_{th}$ **then**
    Compute samples to synthesize: $Q = \beta(m_t - m_s)$, $\beta \in (0, 1]$;
    Normalize the ratio in n-dimensional space for each $x_i \in$ minority class:

$$R_i = \frac{y_i}{\sum_{j=1}^{m_s} \frac{y_j}{k}}$$

    Determine the number of new samples generated for each minority example x_i using $g_i = R_i \times G$, where G is the total number of samples to be synthesized;
    for *each* $x_i \in$ *borderline minority samples* **do**
        Randomly choose minority example x_{ri} from k nearest neighbors;
        Generate new example: $L_i = x_i + \lambda(x_{ri} - x_i)$, $\lambda \in (0, 1]$;
    end
else
    Return original data;
end
Return E;
end

---

density regions and exhibit significant uncertainty, are selected as the most representative attack samples. These high-uncertainty samples are crucial in defining the model's decision boundary. Furthermore, since the primary goal of this study is to detect zero-day attacks, selecting the most representative attack samples significantly reduces the model's reliance on specific known attack samples, thereby improving the discriminator's ability to distinguish between normal and attack data. The formula for density-weighted uncertainty sampling is as follows:

$$U(x_i) = -\left[\psi(x_i^r)\log\left(\psi(x_i^r)\right) + \left(1-\psi(x_i^r)\right)\log\left(1-\psi(x_i^r)\right)\right] + \frac{1}{k}\sum_{j=1}^{k}\exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right) \tag{1}$$

Equation (1) integrates both the uncertainty and density components to compute an overall score, which prioritizes samples exhibiting high uncertainty in dense regions. These selected samples enable the model to effectively learn the decision boundary while mitigating the influence of noisy data, thereby improving the effectiveness of data augmentation. To address the potential exclusion of rare yet informative samples, we introduce a density-weighted strategy. This strategy ensures that, while prioritizing samples from high-density regions, a portion of samples from low-density regions with high uncertainty is also retained. This approach maintains the representation of rare samples, preventing performance degradation in scenarios involving rare attack types. Once representative attack samples near the decision boundary are identified, linear interpolation is employed to generate additional potential anomalous points. This process incrementally shifts the decision boundary toward more challenging regions, enabling the model to adapt to complex sample distributions. By iteratively selecting and augmenting data from both high-density and low-density regions, the resulting diverse and informative samples enrich the training dataset. The augmented data, in conjunction with the original data, is then utilized to train both the generator and the discriminator, thereby resulting in a more robust detection framework. This enhanced framework is capable of effectively addressing both common and rare attack scenarios while also improving the model's generalization capability.

3.2.2. Collaborative focal loss function for discriminators

To address the inherent imbalance challenge in few-shot learning, we introduce a collaborative focal loss function for the discriminators. For both the adversarial classifier $\xi_k (1 \le k \le m)$ and the auxiliary classifier $\varphi_k (1 \le k \le m)$ that compose the discriminator ξ_k, the loss function is partitioned into two components: real data loss $L_r^{D_k}$ and fake data loss $L_f^{D_k}$. The real data loss $L_r^{D_k}$ can be defined as follows:

$$L_r^{D_k} = -\frac{1}{n_r}\sum_{i=1}^{n_r}\left(\left(1-\overline{\xi^{k-1}}\left(y=1 \mid x_i^r\right)\right)^{\beta}\log\xi\left(x_i^r\right)\right. \\ \left. + \left(1-\overline{\varphi^{k-1}}\left(y=y_i^r \mid x_i^r\right)\right)^{\beta}\log\varphi\left(y=y_i^r \mid x_i^r\right)\right) \tag{2}$$

The loss function $L_r^{D_k}$ consists of two main components: the real data loss for the adversarial classifier $-(1-\overline{\xi^{k-1}}(y=1 \mid x_i^r))^{\beta}\log\xi(x_i^r)$ and the real data loss for the auxiliary classifier $-(1-\overline{\varphi^{k-1}}(y=y_i^r \mid x_i^r))^{\beta}\log\varphi(y=y_i^r \mid x_i^r)$. Here, $\beta\in[0,5]$, and when $\beta=0$, the loss function degenerates into a cross-entropy loss function. In addition, $(1-\overline{\xi^{k-1}}(y=1 \mid x_i^r))^{\beta}$ and $(1-\overline{\varphi^{k-1}}(y=y_i^r \mid x_i^r))^{\beta}$ act as regulation factors. $\overline{\xi^{k-1}}(y=1 \mid x_i^r)$ represents the average probability that the current sample x_i^r was classified as real by the previous $k-1$ adversarial classifiers, while $\overline{\varphi^{k-1}}(y=y_i^r \mid x_i^r)$ represents the average probability that the current sample x_i^r was classified as y_i^r by the previous $k-1$ auxiliary classifiers. The purpose of the adjustment factor is to reduce the weight for correctly classified samples while maintaining the weight for uncertain samples, thereby increasing the loss weight for difficult-to-classify samples. Additionally, due to the cooperative nature of each member in the ensemble, when a sample x_i^r is misclassified by a previous member, the loss function assigns a higher weight to this sample, increasing the penalty for misclassification by future members. In unbalanced datasets, these samples are more likely to be targeted. Minimizing the loss function enables the classifier to focus on samples that were previously misclassified, thereby improving overall performance.

Similarly, the false data loss $L_f^{D_k}$ can be defined as:

$$L_f^{D_k} = -\frac{1}{n_f}\sum_{i=1}^{n_f}\left(\left(1-\overline{\xi^{k-1}}\left(y=0 \mid \sum_{j=1}^{N}\pi_j G(z_i^f,y_i^f)\right)\right)^{\beta} \right. \\ \times\log\left(1-\xi\left(\sum_{j=1}^{N}\pi_j G(z_i^f,y_i^f)\right)\right)\Bigg) \\ + \sum_{i=1}^{n_f}\left(\left(1-\overline{\varphi^{k-1}}\left(y=y_i^f \mid \sum_{j=1}^{N}\pi_j G(z_i^f,y_i^f)\right)\right)^{\beta} \right. \\ \left. \times\log\varphi\left(y=y_i^f \mid \sum_{j=1}^{N}\pi_j G(z_i^f,y_i^f)\right)\right) \tag{3}$$

The total loss for the k-th discriminator can be defined as:

$$L^{D_k} = \lambda_1 L_r^{D_k} + (1-\lambda_1)L_f^{D_k} \tag{4}$$

where $\lambda_1\in(0,1)$ is a weighting parameter used to balance the loss functions. By using gradient descent, subsequent discriminators will pay more attention to samples misclassified by previous discriminators, thereby striving to achieve optimal detection outcomes.

3.2.3. Ensemble focal loss function for generators

To enhance the diversity and realism of the samples produced by the generator, and to ensure that these samples adequately cover the boundary regions of the data space, we developed an ensemble focal loss function. This loss function comprises both a boundary loss and adversarial loss, with the objective of optimizing the distribution and realism of the generated samples. Initially, the boundary loss is employed to regulate the distribution of the generator's output, ensuring that the samples not only concentrate within the high-probability regions of the data space but also effectively cover its boundary regions. The boundary loss is defined as follows:

$$L_B^{G_k} = \frac{1}{n_f}\sum_{i=1}^{n_f}\log\left(\left|\alpha - D\left(G\left(z_i^f,y_i^f\right)\right)\right|\right) \\ + \lambda_2\frac{1}{\frac{1}{n_f}\sum_{i=1}^{n_f}\left(\left\|G\left(z_i^f,y_i^f\right)-\mu\right\|_2\right)} \tag{5}$$

The first term, $\log\left(\left|\alpha - D(G(z_i^f,y_i^f))\right|\right)$, serves as a density control mechanism for the generator in the boundary regions. By adjusting the hyperparameter α, the generator can regulate the density of generated samples at the data boundaries, thereby preventing excessive concentration in specific regions. The second term, $\frac{1}{n_f}\sum_{i=1}^{n_f}\left\|G(z_i^f,y_i^f)-\mu\right\|_2$, quantifies the distance between the generated samples and the real data center μ. By weighting the inverse of this distance, the generated samples are encouraged to disperse throughout the entire data space, rather than clustering near the data center. To further refine the generator's performance, we introduced an adversarial loss, derived from feedback provided by the discriminator, to guide the generator in learning a more realistic sample distribution. The adversarial loss is defined as follows:

$$L_C^{G_k} = -\frac{1}{n_f}\sum_{i=1}^{n_f}\left(1-\overline{\xi^{k-1}}\left(y=1 \mid G_i(z_i^f,y_i^f)\right)^{\beta}\right)\log\xi\left(G_i(z_i^f,y_i^f)\right) \\ -\frac{1}{n_f}\sum_{i=1}^{n_f}\left(1-\overline{\varphi^{k-1}}\left(y=y_i^r \mid G_i(z_i^f,y_i^f)\right)^{\beta}\right) \\ \times\log\varphi\left(y=y_i^r \mid G_i(z_i^f,y_i^f)\right) \tag{6}$$

Finally, the boundary loss and adversarial loss associated with G^k are integrated, yielding the total loss for the k-th generator as expressed in equation (7):

$$L_G^k = L_B^{G_k} + \lambda_3\sum_{i=1}^{m}L_C^{G_k} \tag{7}$$

By tuning the hyperparameter λ_3, the influence of the boundary loss and adversarial loss on the generator's total loss is effectively balanced. The application of gradient descent to iteratively optimize the loss function enables the model to eventually converge to the mini-max Nash equilibrium.

3.3. Theoretical analysis

In this section, we provide a theoretical analysis of the improvement in coverage for zero-day attack samples achieved by the generator within the FLCGAN model. On one hand, FLCGAN enhances the model's generalization capability by expanding the feature space of known attacks through attack sample data augmentation techniques, thereby narrowing the distributional gap between known and zero-day attacks. On the other hand, FLCGAN replaces the single generator used in traditional GANs with multiple generators, allowing the generation of multiple reference distributions across the entire dataset. This approach increases the coverage of generated adversarial samples, thereby enhancing the diversity of generated samples. A detailed theoretical analysis follows.

3.3.1. Improving coverage by data augmentation

We propose a data augmentation method based on uncertainty sampling with density weighting, aimed at reinforcing samples with high uncertainty and high density. To assess the effectiveness of this approach, we employ the Wasserstein distance [38] as the evaluation criterion. The augmented distribution $P_{\text{augmented}}$ spans a broader feature space, particularly covering regions near the decision boundary, which are the most difficult to address during model training and most likely to contain features of zero-day attacks. As $P_{\text{augmented}}$ encompasses a large portion of the feature spaces, the distance between it and the zero-day attack distribution $P_{\text{zero-day}}$ is consequently reduced. Specifically, the Wasserstein distance decreases as follows: $W(P_{\text{augmented}}, P_{\text{zero-day}}) < W(P_{\text{know}}, P_{\text{zero-day}})$. Therefore, this data augmentation method expands the feature space, narrows the distributional gap between known and zero-day attacks, and enhances the model's generalization ability by increasing the distributional similarity between known attack data and zero-day attack data.

3.3.2. Improving coverage by multi-generator and multi-discriminator

The multi-generator and multi-discriminator structure of FLCGAN enables the generated adversarial samples to more effectively cover a wider range of zero-day attack scenarios. A detailed theoretical analysis is presented below.

In a generative adversarial network, the generator G and the discriminator D participate in the following mini-max game:

$$\min_{G}\max_{D}\mathcal{J}(G,D)=\mathbb{E}_{\mathbf{x}\sim P_{\text{data}}}(\log D(\mathbf{X}))+\mathbb{E}_{\mathbf{x}\sim P_{\text{model}}}(\log(1-D(G(\mathbf{X})))) \tag{8}$$

For convenience, we introduce an intermediate variable $C_j(\mathbf{X})$ to represent the probability generated by G_j, along with a hyperparameter $\varepsilon>0$ to balance the intermediate variable. Consequently, the equation can be reformulated as:

$$\min_{G,C}\max_{D}\mathcal{J}(G,D)=\mathbb{E}_{\mathbf{x}\sim P_{\text{data}}}(\log D(\mathbf{X}))+\mathbb{E}_{\mathbf{x}\sim P_{\text{model}}}(\log(1-D(G(\mathbf{X}))))$$
$$-\varepsilon\sum_{j=1}^{N}\pi_j\mathbb{E}_{\mathbf{x}\sim P_{G_j}}\left(\log C_j(\mathbf{X})\right) \tag{9}$$

To obtain the Nash equilibrium solution of the mini-max game in Equation (9), we assume that the weights associated with the multiple generators $G_{1:N}$ in the CGAN are $\pi_1,\pi_2,\dots,\pi_N$. As a result, we can derive the following expression:

$$\begin{aligned}\frac{\delta\mathcal{J}}{\delta C_j(\mathcal{X})}&=-\varepsilon\frac{\delta}{\delta C_j(\mathcal{X})}\int\left(\pi_1P_{G_1}(\mathcal{X})\log\left(1-\sum_{j=2}^{N}C_j(\mathcal{X})\right)\right.\\&\quad\left.+\sum_{j=2}^{N}\pi_jP_{G_j}(\mathcal{X})\log C_j(\mathcal{X})\right)d\mathcal{X}\\&=-\varepsilon\left(\frac{\pi_kP_{G_k}(\mathcal{X})}{C_j(\mathcal{X})}-\frac{\pi_1P_{G_1}(\mathcal{X})}{C_1(\mathcal{X})}\right)\end{aligned} \tag{10}$$

By setting the first-order partial derivative $\frac{\delta\mathcal{J}}{\delta C_j(\mathcal{X})}=0$, and substituting $j\in\{2,\dots,N\}$, the following expression is obtained:

$$\frac{\pi_1P_{G_1}(\mathcal{X})}{C_1^*(\mathcal{X})}=\frac{\pi_2P_{G_2}(\mathcal{X})}{C_2^*(\mathcal{X})}=\cdots=\frac{\pi_NP_{G_N}(\mathcal{X})}{C_N^*(\mathcal{X})} \tag{11}$$

Additionally, since $\sum_{j=1}^{N}C_j^*(\mathcal{X})=1$, the optimal $C_j^*(\mathcal{X})$ at the Nash equilibrium point in Equation (9) can be expressed as:

$$C_j^*(\mathcal{X})=\frac{\pi_jP_{G_j}(\mathcal{X})}{\sum_{j=1}^{N}\pi_jP_{G_j}(\mathcal{X})} \tag{12}$$

Moreover, as demonstrated in Reference [39], the optimal D^* at the Nash equilibrium point specified in Equation (9) can be expressed as:

$$D^*=\frac{P_{\text{data}}(\mathcal{X})}{P_{\text{data}}(\mathcal{X})+P_{\text{model}}(\mathcal{X})} \tag{13}$$

To derive the expression for the optimal G^* at the Nash equilibrium point, we substitute the optimal D^* and $C_j^*(\mathcal{X})$ into Equation (9). By reorganizing the equation in terms of $G_{1:N}$, we obtain:

$$\begin{aligned}\mathcal{L}(G_{1:N})=&\mathbb{E}_{\mathcal{X}\sim P_{\text{data}}}\left(\log\frac{P_{\text{data}}(\mathcal{X})}{P_{\text{data}}(\mathcal{X})+P_{\text{model}}(\mathcal{X})}\right)\\&+\mathbb{E}_{\mathcal{X}\sim P_{\text{model}}}\left(\log\frac{P_{\text{model}}(\mathcal{X})}{P_{\text{data}}(\mathcal{X})+P_{\text{model}}(\mathcal{X})}\right)\\&-\varepsilon\sum_{j=1}^{N}\pi_j\mathbb{E}_{\mathcal{X}\sim P_{G_j}}\left(\log\frac{\pi_jP_{G_j}(\mathcal{X})}{\sum_{k=1}^{N}\pi_kP_{G_k}(\mathcal{X})}\right)\\&=2\text{JSD}(P_{\text{data}}\parallel P_{\text{model}})-\varepsilon\text{JSD}_{\pi}(P_{G_{1:N}})\\&-2\log 2-\varepsilon\sum_{j=1}^{N}\pi_j\log\pi_j\end{aligned} \tag{14}$$

Consequently, at the Nash equilibrium point in Equation (9), the optimal generator G^* can be expressed as:

$$G^*=\arg\min_{G}\left(2*\text{JSD}(P_{\text{data}}\parallel P_{\text{model}})-\varepsilon\text{JSD}_{\pi}(P_{G_{1:N}})\right) \tag{15}$$

Thus, it can be concluded that at the Nash equilibrium of the mini-max game, the combination of multiple reference distributions produced by the generators $G^*=[G_1^*,G_2^*,\dots,G_N^*]$ can be represented as $P^*_{\text{model}}(\mathcal{X})$:

$$P^*_{\text{model}}(\mathcal{X})=\sum_{j=1}^{N}\pi_jP^*_{G_j}(\mathcal{X})\approx P_{\text{data}}(\mathcal{X}) \tag{16}$$

We assume that the distribution of zero-day attack data is represented by $P_{\text{Zero-Day}}(\mathcal{X})$, while the distribution of known attack data is denoted as

$$P_{\text{data}}(\mathcal{X})=P_{\text{known}}(X_1)\cup P_{\text{known}}(X_2)\cup\dots\cup P_{\text{known}}(X_n).$$

The Jensen-Shannon divergence, denoted as $\text{JSD}(P_{\text{Zero-Day}}(\mathcal{X})\parallel P_{\text{data}}(\mathcal{X}))=\mathcal{K}(\mathcal{K}\in[0,1])$, quantifies the similarity between these two distributions. Consequently, $P^*_{\text{model}}(\mathcal{X})$ can be expressed as:

$$P^*_{\text{model}}(\mathcal{X})=\sum_{j=1}^{N}\pi_jP^*_{G_j}(\mathcal{X})\approx P_{\text{data}}\begin{cases}\approx P_{\text{Zero-Day}}(\mathcal{X})(\mathcal{K}\to 0)\\\neq P_{\text{Zero-Day}}(\mathcal{X})(\mathcal{K}\to 1)\end{cases} \tag{17}$$

Based on Equation (17), it can be further deduced that the model coverage $P^*_{\text{model}}(\mathcal{X})$ in the FLCGAN model is contingent upon the similarity $\mathcal{K}$. The closer the zero-day attack data distribution $P_{\text{Zero-Day}}(\mathcal{X})$ is to the known attack data distribution $P_{\text{data}}(\mathcal{X})$ (as $\mathcal{K}\to 0$), the more likely it is that $P^*_{\text{model}}(\mathcal{X})$ will generate potential sample points near the zero-day attack data distribution $P_{\text{Zero-Day}}(\mathcal{X})$.

Simultaneously, this process can effectively mitigate the risk of the generator being biased towards a single reference distribution $P_{G_j}(\mathcal{X})$, thus avoiding the problem of failing to fully capture the entire range of the zero-day attack distribution $P_{\text{Zero-Day}}(\mathcal{X})$.

4. Experiment

In this section, we provide a comprehensive description of the experimental setup for FLCGAN, including the simulation tools and datasets utilized, model evaluation metrics, benchmark models, and model parameter configurations. Additionally, ablation experiments were conducted to assess the contributions of different components. Specifically, four types of attacks were selected for the experiments: DoS, RandomSpeed, Disruptive, and RandomPosOffset. During the training phase, three of these attack types were treated as known attacks and combined with normal data to construct the training set. In the testing phase, the remaining attack type, which was excluded from the training data, was used as a zero-day attack to evaluate the model's detection performance and generalization ability.

Table 1
F2MD simulation parameters.

Simulation parameters	Value
Road network topology	LuSTNanoScenario
Simulation time	5 min
Pseudonym Change Protocol	Car2Car
Vehicle motion model	Random
Communication protocol	IEEE 802.11p/1609
Attack vehicle density	1% 3% 5% 10%
Attack types	DoS RS RPO Disruptive
Beacon transmission frequency	1 Hz
Obstacle decay model	Simple

Table 2
Description of attack types in the dataset.

Attack Types	Description
DoS	The attacker has the capability to augment the beaconing frequency by a specific factor and has the discretion to transmit legitimate messages, arbitrary data, or opt to frequently modify and substitute preloaded kana
RS	The vehicle broadcasts a random speed to deceive others, within a range $(0 \rightarrow V_{\max})$
RPO	The vehicle broadcasts its real position with a random offset limited to a max value $\Delta(0 \rightarrow V_{\max})$
Disruptive	The attacker chooses a random beacon from the received history and replays its data, aiming to disrupt the system by flooding the network with old beacons data

4.1. Experimental setup

4.1.1. Simulation tools and datasets

To assess the efficacy of the proposed method, the IoV simulation tool Framework For Misbehavior Detection (F2MD) was utilized to generate the requisite datasets [40]. In comparison to the publicly available VeReMi dataset [41], F2MD offers a more extensive and detailed collection of IoV attack types. The primary simulation parameters of F2MD are presented in Table 1.

In F2MD, we systematically collected and organized the generated Basic Safety Messages (BSMs) into comprehensive datasets. BSMs serve as the primary units of information transmission within IoV environment, recording the actual road conditions of both normal and attack vehicles. Within F2MD, we implemented four distinct types of network attacks, including DoS, RandomSpeed (RS), RandomPosOffset (RPO), and Disruptive attacks. The RS and RPO attacks involve data tampering, while Disruptive attacks are executed through data injection. The DoS attack is an amalgamation of RS and RPO strategies. Each of these attacks varies in its potential impacts and severity. By implementing these scenarios, we can rigorously evaluate the performance and resilience of the proposed method under diverse attack conditions. Detailed descriptions of these network attacks are provided in Table 2.

4.1.2. Comparative methods and parameter settings

To assess the effectiveness of the proposed FLCGAN method, it was benchmarked against eleven existing methods, utilizing F1 score and AUC value as performance metrics. These eleven methodologies are classified into four categories: (i) a SVM-based method, OCSVM; (ii) an ensemble detector, iForest; (iii) six deep learning detection methods, namely DSVDD [19], ICL [42], DAGMM [43], DSAD [44], PreNet [45], and SLAD [46]; and (iv) three GAN-based methods, specifically FGAN [47], MOAL [24], and EAL-GAN [14].

The FLCGAN model was configured with the following parameters: (i) Two generators and five discriminators were employed. (ii) The generator was structured as a three-layer neural network $(128 * 2d * d)$, with weights $\pi_1, \pi_2, ..., \pi_N$ initialized as $\frac{1}{N}$. The discriminator was designed as a $(d * 2d * 2)$ neural network, where 128 represents the input data dimension obtained by concatenating the input noise z_i^f and class label y_i^f, and d denotes the feature dimension of the data. The adversarial and auxiliary classifier within the discriminator share the first two layers of the network, with each classifier producing a one-dimensional output. (iii) All generators and discriminators were initialized using an orthogonal initializer, and spectral normalization was applied to normalize the weight matrix of each fully connected layer, ensuring a maximum singular value of 1. (iv) The anomaly threshold τ was set to 0.5. (v) The Adam optimizer was used, with the learning rate set to 0.01 for both generator and discriminator, and the initial vectors were orthogonal. (vi) The mini-batch size for training was $m = 128$, and the maximum number of training epochs was set to 80. (vii) In each iteration, 20% of the most representative samples were selected for data augmentation. (viii) In the loss function, the parameters were set as $\lambda_1 = 0.25$, $\lambda_2 = 0.2$, $\lambda_3 = 1$, $\alpha = 0.5$, and $\beta = 2$.

4.1.3. Evaluation measures

To comprehensively evaluate the performance of the proposed method, the F1 score was chosen as the primary evaluation metric. The F1 score effectively combines precision and recall, offering a balanced measure of the model's performance. Additionally, the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) were employed to further assess the model's detection accuracy. The ROC curve illustrates the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) across various threshold settings, while the AUC value provides an aggregate measures of the classifier's effectiveness. Since both the F1 score and AUC value are robust to class imbalance, they serve as reliable indicators of the model's overall performance. As a result, these two metrics were selected to assess the efficacy of the proposed zero-day attack detection method.

4.2. Experimental results

In this subsection, we first compare FLCGAN with eight machine learning and deep learning-based methods, followed by a comparative analysis with three GAN-based approaches. The performance of these methods, evaluated in terms of AUC and F1 scores, is presented in Fig. 3 and Table 3, respectively.

According to the data in Fig. 3, when AUC is used as the evaluation metric, the proposed FLCGAN method outperforms detection methods such as ICL (82.90), DAGMM (92.63), and DSVDD (82.23). This suggests that certain algorithms with lower AUC values were unable to effectively exploit the characteristics of known attacks for zero-day attack detection. Specifically, when attack data exhibit significant feature variations, these algorithms struggle to capture the key distinguishing features, leading to suboptimal detection of zero-day attacks. Additionally, when evaluated using the F1 score, FLCGAN also demonstrates superior performance, surpassing other methods in this metric. The comprehensive experimental results reveal that FLCGAN consistently achieves better overall performance compared to other approaches, particularly in balancing precision and recall, as reflected in its F1 score.

Furthermore, to provide a more comprehensive comparison of FLCGAN with other GAN-based methods, we conducted extensive comparative experiments and evaluated their detection performance using multiple evaluation metrics. The results of the four methods, assessed across these metrics, are presented in Table 4. As demonstrated in the table, FLCGAN consistently delivers strong performance across all evaluated metrics, further highlighting its effectiveness in detecting zero-day attacks.

To validate the generalizability of the proposed method, we further tested it using the UNSW-NB15 IoT intrusion detection dataset [48] and conducted comparative experiments with two other IoT attack detection methods, IGRF-RFE [49] and CNN-BiLSTM [50]. The experimental results are depicted in Fig. 4. As illustrated, the proposed method slightly outperforms the other two methods in terms of overall performance, notably demonstrating a significant reduction in the false positive rate.

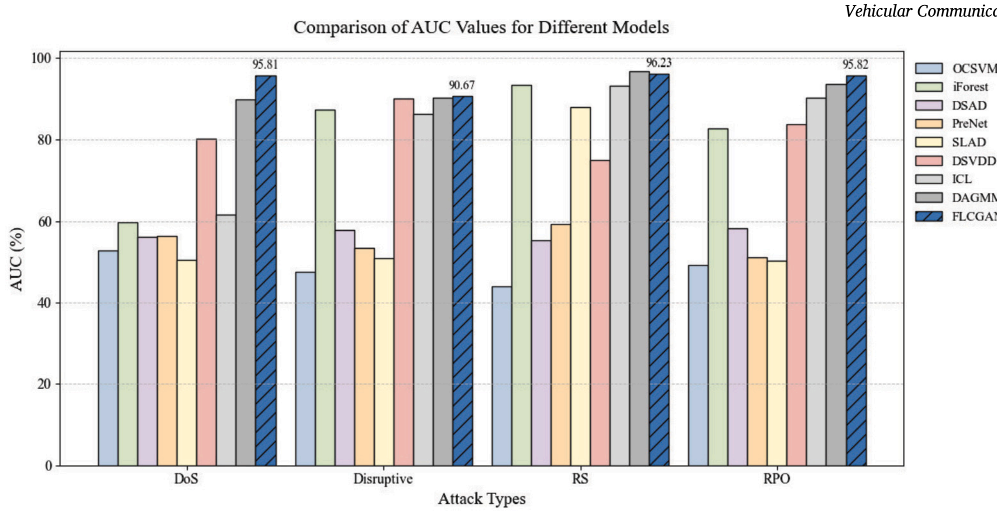

Fig. 3. Comparison of AUC values for different models across various attack types.

Table 3
Comparison of FLCGAN with the remaining 8 models in terms of average F1 values.

Attack Types	OCSVM	iForest	DSAD	PreNet	SLAD	DSVDD	ICL	DAGMM	FLCGAN
DoS	56.62	36.48	65.37	57.05	65.19	79.30	42.86	79.82	**91.34**
Disruptive	45.64	86.03	60.53	67.57	68.02	82.35	85.59	78.07	**90.68**
RS	41.67	92.33	61.76	67.75	88.51	72.09	93.23	94.31	**95.98**
RPO	45.96	80.54	62.87	59.70	66.67	77.66	90.32	86.67	**95.28**
Average	47.47	73.85	62.63	63.02	72.10	77.85	78.00	84.72	**93.32**
Rank	8.5	4.75	6.75	7	5.5	4.5	3.75	4	**1**

Table 4
Multi-metrics Comparative Experiment.

Attack Type	Model	AUC	F1	G-Mean	K-S	MCC	AUPRC
DoS	FGAN	81.25	65.61	51.69	45.39	38.11	63.72
	MOGL	85.61	82.18	82.39	67.20	64.80	86.98
	EAL-GAN	89.65	84.85	83.52	68.36	70.58	88.76
	FLCGAN	**95.81**	**91.34**	**86.32**	**69.52**	**71.09**	**90.92**
Disruptive	FGAN	86.35	85.99	73.26	70.41	63.86	77.23
	MOGL	84.37	81.19	80.35	62.40	60.89	90.82
	EAL-GAN	88.63	86.65	83.59	77.54	77.54	92.75
	FLCGAN	**90.67**	**90.68**	**88.60**	**88.60**	**78.97**	**94.88**
RS	FGAN	86.24	85.56	92.99	88.20	88.03	92.22
	MOGL	91.48	91.19	91.56	89.60	86.88	94.53
	EAL-GAN	93.58	93.52	91.85	93.60	94.58	95.28
	FLCGAN	**96.23**	**95.98**	**93.85**	**93.80**	**90.82**	**96.35**
RPO	FGAN	84.21	80.52	80.79	81.62	89.23	91.16
	MOGL	88.34	85.80	85.04	85.08	85.08	93.14
	EAL-GAN	93.92	91.67	90.28	90.67	85.68	93.65
	FLCGAN	**95.82**	**95.28**	**92.97**	**91.85**	**87.27**	**96.45**

These results indicate that the proposed method exhibits strong generalization capability, effectively transitioning from IoV domain to the IoT domain while maintaining reliable detection accuracy.

To assess the robustness of the proposed method against variations in attack density, we employed the Disruptive attack, which had proven more difficult to detect in prior experiments, and compared the detection performance across different attack densities. The experimental results are presented in Fig. 5. As shown, although the overall AUC and F1 scores of the proposed method experience a slight decrease as the attack density lowers, the AUC consistently remains above 0.90, and the F1 score stays above 0.88. These results suggest that the method is particularly effective at detecting various attacks when the proportion of attack samples in the training set is higher. However, even in scenarios with minimal samples, such as when the attack density is as low as 1%, FLCGAN still achieves relatively strong detection performance, demonstrating its ability to handle highly imbalanced datasets effectively.

4.3. Ablation experiment

We also performed ablation experiments on FLCGAN and its variants to highlight the effectiveness of FLCGAN in three critical areas: the use of multiple generators and multiple discriminators network framework, the application of attack sample augmentation based on either entropy selection or random sampling, and the choice of the ensemble focal loss function and the collaborative focal loss function. The results of these experiments are detailed in Table 5.

As shown in Table 5, FLCGAN achieved the highest F1 scores and average F1 scores across four different datasets, outperforming the

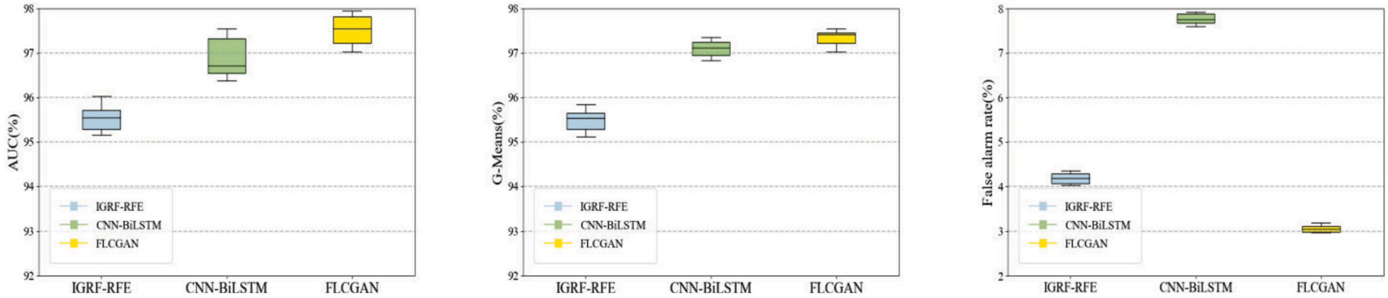

Fig. 4. Comparative experiments on the UNSW-NB15 dataset.

Table 5
Comparison among FLCGAN variants.

Ablation Settings			Attack Types				Avg F1 (as %)
Multi Generators and Discriminators	Data Selection & Augmentation	Focal Loss	DoS	Disruptive	RS	RPO	
×	×	×	81.16±3.51	85.18±2.42	84.16±3.89	88.38±6.82	84.72±4.16
✓	×	×	88.42±1.26	87.98±1.51	91.77±3.18	93.36±1.58	90.38±1.88
✓	✓	×	88.91±1.02	88.32±0.48	92.02±2.47	93.98±0.34	90.81±1.05
✓	✓	✓	**91.34±0.68**	**90.68±0.33**	**95.98±0.25**	**95.28±0.26**	**93.32±0.38**

Table 6
Comparison of the performance of single and multiple generators.

Model Architecture	AUC (%)	F1 (%)	Detection Delay (ms)
Single Generator	91.43	89.87	0.04
Multiple Generators	**94.63**	**93.32**	**0.04**

Table 7
Comparison of detection results with data enhancement using FLCGAN and F2MD.

Attack Type	FLCGAN-F1 (%)	F2MD-F1 (%)	FLCGAN-AUC (%)	F2MD-AUC (%)
DoS	91.34	83.42	95.81	85.43
Disruptive	90.68	84.96	90.67	84.47
RS	95.98	85.32	96.23	87.92
RPO	95.28	87.68	95.82	87.16
Average	93.32	85.35	94.63	86.25

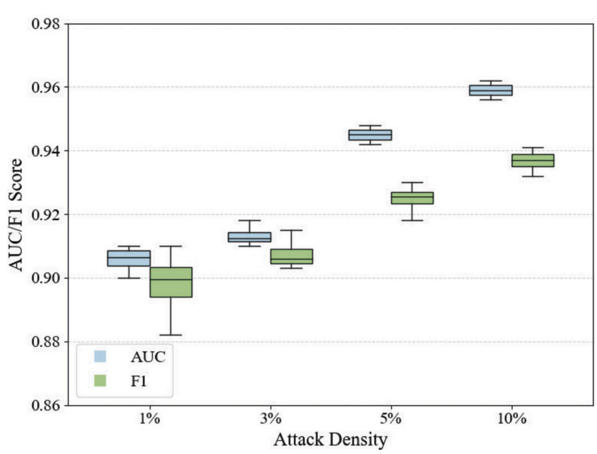

Fig. 5. Impact of attack density on FLCGAN model.

other three variants, particularly excelling in the RS and RPO datasets with F1 scores of 95.98% and 95.28%, respectively. This highlights the critical role of each component, including multi-generators, discriminators, and attack sample data augmentation, in enhancing the model's overall performance. Furthermore, as presented in Table 6, the multiple-generator framework significantly outperforms the single-generator framework. Specifically, the multiple-generator framework achieves an AUC of 94.63% and an F1 score of 93.32%, compared to an AUC of 91.43% and an F1 score of 89.87% obtained by the single-generator framework. Notably, both frameworks exhibit the same detection delay of 0.04 ms, demonstrating that the computational efficiency remains unaffected. This is primarily attributed to the parallel design of the multiple-generator architecture. These results validate the efficacy of the proposed framework in enhancing detection accuracy while maintaining the low latency required for real-time zero-day attack detection in IoV applications. To assess the quality of the data generated by FLCGAN for zero-day attack detection, we compared its performance with data generated by the F2MD simulator across four attack types, as shown in Table 7. The results demonstrate that FLCGAN-generated data consistently outperforms F2MD-generated data, achieving an average F1 score of 93.32% and an average AUC of 94.63%, compared to the F2MD simulator's 85.35% and 86.25%, respectively. This highlights FLCGAN's ability to generate diverse and representative data that more effectively captures the characteristics of zero-day attacks, particularly in scenarios where F2MD-generated data struggles to generalize.

In this study, the attack sample data augmentation method optimized input samples for the FLCGAN model by selecting the most representative attack samples to filter out noisy data. To further investigate the impact of varying data augmentation ratios on model performance, we

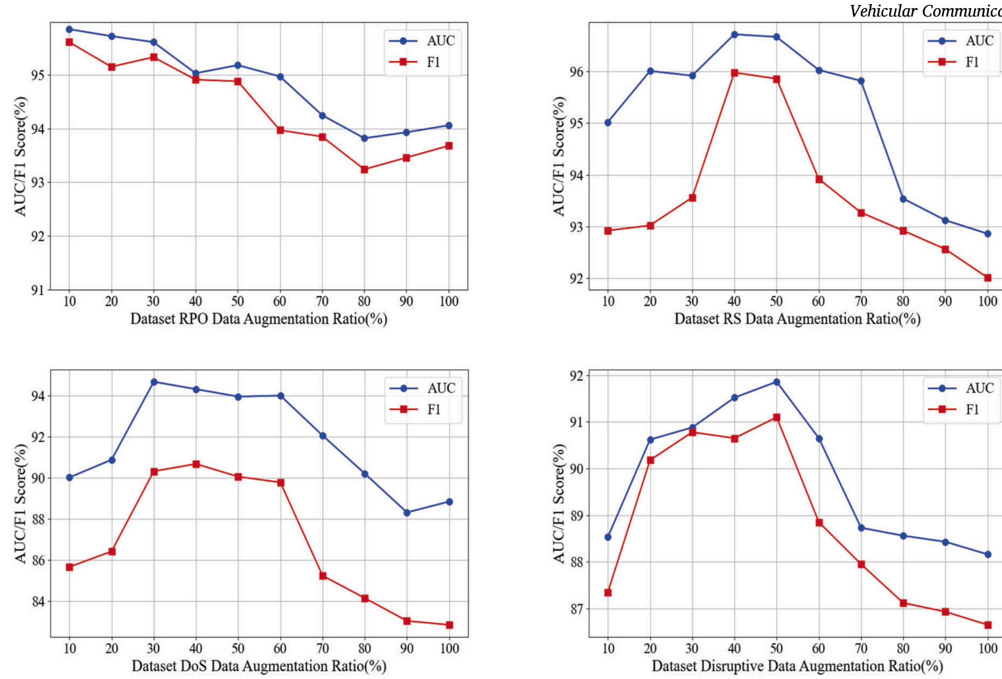

Fig. 6. Impact of data augmentation ratio on FLCGAN model.

Table 8
Model performance under different definitions of loss function.

Metric	DoS		Disruptive		RS		RPO	
	AUC	F1	AUC	F1	AUC	F1	AUC	F1
CE	88.32	84.62	88.72	86.45	91.24	87.87	95.35	92.68
WD	83.92	78.35	77.75	64.82	81.76	66.36	67.85	62.35
KL	71.38	66.98	88.83	71.76	**96.25**	87.76	88.76	85.81
L2	72.15	68.75	88.92	70.76	93.28	66.43	83.78	80.86
Ours	**95.81**	**91.34**	**90.67**	**90.68**	96.23	**95.98**	**95.82**	**95.28**

conducted experiments across four datasets using different data augmentation ratios, with AUC and F1 scores as the evaluation metrics. The results of these experiments are presented in Fig. 6.

The results demonstrate that the model achieves optimal AUC and F1 scores when the data augmentation ratio is between 30% and 50%. This range achieves an optimal balance between expanding the feature space and preserving the original data distribution. At lower augmentation ratios (e.g., 10% to 20%), the number of generated samples is insufficient to significantly enhance the diversity of features, limiting the model's ability to generalize to unseen attack scenarios. Conversely, when the augmentation ratio exceeds 50%, the excessive number of generated samples diminishes the relevance of the original data, potentially introducing noise and bias to the data distribution, which leads to a reduction in model performance. Therefore, an augmentation ratio within the range of 30% to 50% effectively covers potential decision boundaries while preserving the original data characteristics, thereby significantly enhancing the model's detection capability and robustness.

Additionally, we conducted ablation experiments on the FLCGAN model, comparing the ensemble focal loss and collaborative focal loss functions (ours) with four other loss functions: Cross-Entropy Loss (CE), Wasserstein Distance (WD), Kullback-Leibler Divergence (KL), and Euclidean Distance (L2). The experimental results are presented in Table 8. The results indicate that the proposed loss functions achieved superior detection performance compared to the other four. As shown in the table, the Cross-Entropy Loss function performed well on certain datasets. However, the Wasserstein distance proved unsuitable as a loss function for FLCGAN because it accounts for distribution shift, which can introduce noise issues. Additionally, the non-symmetry of the KL Divergence and the sensitivity of the L2 distance to minor variations make them less effective in meeting the FLCGAN model's requirements.

4.4. Detection delay

Standardized vehicle communication mandates that vehicle safety services comply to stringent performance standards, which are primarily evaluated based on two key metrics: packet delivery rate and detection latency. In the development of attack detection systems, the latency metric is particularly crucial to satisfy the real-time requirements of IoV. Latency refers to the time it take for a data packet to travel from its source to its destination. According to the U.S. Department of Transportation, the maximum allowable latency for the high-priority vehicle safety services, such as collision and attack warnings, must range between 10 and 100 milliseconds [51]. In contrast, vehicle traffic safety applications, such as autonomous or cooperative driving, impose even stricter latency requirements, typically ranging from 10 to 20 milliseconds [52]. As a result, to meet the stringent real-time or latency constraints of vehicle communication, the processing time for each network packet should be less than 10 milliseconds.

In this section, we selected the detection models ICL and DAGMM, which demonstrated strong performance in Section 4.2, and compared them with the proposed FLCGAN to evaluate detection latency. Fig. 7 presents a comparison of the detection latency among these three methods. As depicted in the figure, the detection latency of the proposed FLCGAN is approximately three times faster than that of the DAGMM

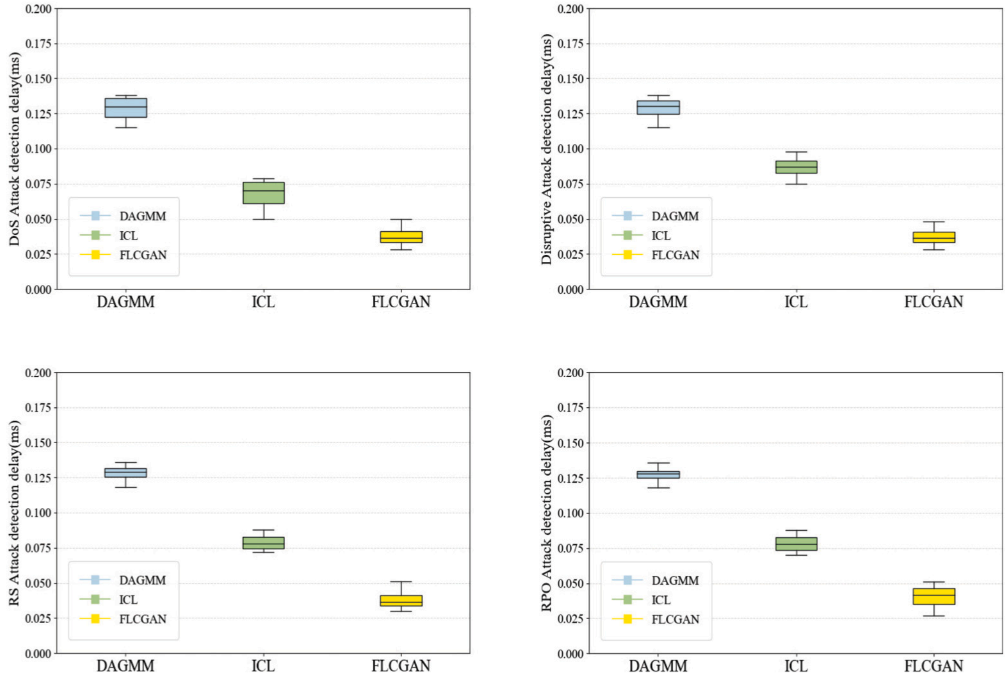

Fig. 7. Attack detection delay.

method and about 1.5 times faster than the ICL method, with an average detection time per data packet of approximately 0.04 milliseconds, well below the 10-millisecond requirement for IoV security latency. Significance analysis revealed that the p-value of the observed difference between FLCGAN and the other two methods was less than 0.0001.

On one hand, even when accounting for communication and network latency, FLCGAN's performance remains far superior to that of DAGMM, primarily because FLCGAN relies solely on the auxiliary classifier for detection, thereby significantly reducing the complex forward propagation process typical of traditional methods like DAGMM. Consequently, FLCGAN is better suitable for latency-constrained applications, such as attack detection in IoV. On the other hand, although the ICL method achieved relatively good detection latency, its lower AUC and F1 scores indicate that it may not perform effectively in more complex attack detection scenarios. Therefore, the proposed FLCGAN method proves to be more suitable for attack detection in vehicular environments.

5. Discussion

In this paper, we validated the effectiveness of the proposed zero-day attack detection method for the Internet of Vehicles (IoV). Compared to existing methods, our approach demonstrates superior performance in terms of detection rate, false positive rate, and detection latency. These results highlight the significant potential of the proposed method for practical applications in intelligent transportation systems and autonomous driving systems.

The experimental results highlight two key contributions of this method. First, the attack sample data augmentation method effectively expands the feature space, enhancing the detection model's generalization capability in unknown attack scenarios while significantly reducing the false positive rate. Second, the ensemble focal loss function fosters greater diversity in the generated zero-day attack samples, while the collaborative focal loss function improves the classification accuracy for hard-to-classify samples, thereby addressing the issue of data imbalance. Collectively, these components enhance the robustness and accuracy of the detection framework, ensuring its effectiveness in IoV scenarios where labeled attack samples are limited.

While the proposed method demonstrates strong performance across multiple evaluation metrics, it does have certain limitations. Although zero-day attacks in the IoV are assumed to share similarities with known attack types, challenges remain when addressing the evolving nature and variants of zero-day attacks. If these zero-day attacks diverge significantly from the patterns of known attacks, the detection effectiveness of the proposed method may not achieve substantial improvements over existing approaches. Consequently, future research could explore the integration of online learning methods to continuously update the database of known attacks, thereby enhancing the diversity of recognized attacks and improving the detection capability for emerging zero-day attacks.

6. Conclusions

To address the issue of high false positive rates commonly associated with anomaly-based zero-day attack detection methods within the IoV domain, this paper introduces a few-shot learning-based conditional generative adversarial network (FLCGAN) for zero-day attack detection in IoV. The proposed model employs multiple generators and discriminators to generate potential zero-day attack data from known attack samples, thereby enhancing the model's ability to capture novel features characteristic of zero-day attacks. An attack sample data augmentation algorithm has been developed to optimize the input samples for FLCGAN, contributing to a reduction in false positive rates. To tackle the challenge of data imbalance in the generated samples, an ensemble focal loss function is applied to the generators, while a collaborative focal loss function is utilized for the discriminators, promoting collaboration among them to more effectively identify difficult-to-classify data. Extensive experiments conducted on the F2MD IoV simulation platform demonstrate that the proposed approach not only achieves superior detection performance but also reduced detection latency for zero-day attacks in IoV. Future work will explore the incorporation of an adaptive parameter adjustment mechanism that dynamically tunes model parameters based on real-time data feedback, further enhancing detection performance.

CRediT authorship contribution statement

Bingfeng Xu: Writing – review & editing, Writing – original draft, Validation, Methodology, Funding acquisition, Data curation, Conceptualization. **Jincheng Zhao:** Writing – review & editing, Writing – original draft, Validation, Methodology. **Bo Wang:** Writing – review & editing, Validation, Investigation, Conceptualization. **Gaofeng He:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by National Natural Science Foundation of China under grants 62372240 and 62372239, by Jiangsu Provincial Key Laboratory of Network and Information Security under Grants No. BM2003201.

Data availability

Data will be made available on request.

References

[1] M. Xu, D.T. Hoang, J. Kang, D. Niyato, Q. Yan, D.I. Kim, Secure and reliable transfer learning framework for 6g-enabled internet of vehicles, IEEE Wirel. Commun. 29 (4) (2022) 132–139.
[2] D.P.M. Osorio, I. Ahmad, J.D.V. Sánchez, A. Gurtov, J. Scholliers, M. Kutila, P. Porambage, Towards 6g-enabled internet of vehicles: security and privacy, IEEE Open J. Comput. Soc. 3 (2022) 82–105.
[3] R. Deemantha, B. Hettige, Autonomous car: current issues, challenges and solution: a review, in: 15th International Research Conference, 2023, pp. 1–7.
[4] Sergiu Gatlan, Tesla hacked, 24 zero-days demoed at pwn2own automotive 2024, available at https://www.bleepingcomputer.com/news/security/tesla-hacked-24-zero-days-demoed-at-pwn2own-automotive-2024/, 2024. (Accessed 24 January 2024).
[5] Y. Guo, A review of machine learning-based zero-day attack detection: challenges and future directions, Comput. Commun. 198 (2023) 175–185.
[6] K. Wang, A. Zhang, H. Sun, B. Wang, Analysis of recent deep-learning-based intrusion detection methods for in-vehicle network, IEEE Trans. Intell. Transp. Syst. 24 (2) (2022) 1843–1854.
[7] E. Hotellier, F. Sicard, J. Francq, S. Mocanu, Standard specification-based intrusion detection for hierarchical industrial control systems, Inf. Sci. 659 (2024) 120102.
[8] L. Yang, A. Moubayed, A. Shami, Mth-ids: a multitiered hybrid intrusion detection system for internet of vehicles, IEEE Internet Things J. 9 (1) (2021) 616–632.
[9] Y. Liu, H. Xue, W. Zhuang, F. Wang, L. Xu, G. Yin, Ct2-mds: cooperative trust-aware tolerant misbehaviour detection system for connected and automated vehicles, IET Intell. Transp. Syst. 16 (2) (2022) 218–231.
[10] J. Zhao, S. Shetty, J.W. Pan, C. Kamhoua, K. Kwiat, Transfer learning for detecting unknown network attacks, EURASIP J. Inf. Secur. 2019 (1) (2019) 1–13.
[11] M. Thankappan, H. Rifà-Pous, C. Garrigues, Multi-channel man-in-the-middle attacks against protected wi-fi networks: a state of the art review, Expert Syst. Appl. 210 (2022) 118401.
[12] N. Sameera, M. Shashi, Deep transductive transfer learning framework for zero-day attack detection, ICT Express 6 (4) (2020) 361–367.
[13] S.T. Mehedi, A. Anwar, Z. Rahman, K. Ahmed, R. Islam, Dependable intrusion detection system for iot: a deep transfer learning based approach, IEEE Trans. Ind. Inform. 19 (1) (2022) 1006–1017.
[14] Z. Chen, J. Duan, L. Kang, G. Qiu, Supervised anomaly detection via conditional generative adversarial network and ensemble active learning, IEEE Trans. Pattern Anal. Mach. Intell. 45 (6) (2022) 7781–7798.
[15] J. Yu, H. Do, Proximity-based density description with regularized reconstruction algorithm for anomaly detection, Inf. Sci. 654 (2024) 119816.
[16] I. Nouretdinov, J. Gammerman, M. Fontana, D. Rehal, Multi-level conformal clustering: a distribution-free technique for clustering and anomaly detection, Neurocomputing 397 (2020) 279–291.
[17] J. Yang, Y. Chen, S. Rahardja, Neighborhood representative for improving outlier detectors, Inf. Sci. 625 (2023) 192–205.
[18] S. Wang, X. Hu, J. Sun, J. Liu, Hyperspectral anomaly detection using ensemble and robust collaborative representation, Inf. Sci. 624 (2023) 748–760.
[19] L. Ruff, R. Vandermeulen, N. Goernitz, L. Deecke, S.A. Siddiqui, A. Binder, E. Müller, M. Kloft, Deep one-class classification, in: International Conference on Machine Learning, PMLR, 2018, pp. 4393–4402.
[20] X. Zhao, Y. Wu, D.L. Lee, W. Cui, iforest: interpreting random forests via visual analytics, IEEE Trans. Vis. Comput. Graph. 25 (1) (2018) 407–416.
[21] H. Xu, G. Pang, Y. Wang, Y. Wang, Deep isolation forest for anomaly detection, IEEE Trans. Knowl. Data Eng. 35 (12) (2023) 12591–12604.
[22] D. Yang, M. Hwang, Unsupervised and ensemble-based anomaly detection method for network security, in: 2022 14th International Conference on Knowledge and Smart Technology (KST), IEEE, 2022, pp. 75–79.
[23] H. Kye, M. Kim, M. Kwon, Hierarchical autoencoder for network intrusion detection, in: ICC 2022-IEEE International Conference on Communications, IEEE, 2022, pp. 2700–2705.
[24] Y. Liu, Z. Li, C. Zhou, Y. Jiang, J. Sun, M. Wang, X. He, Generative adversarial active learning for unsupervised outlier detection, IEEE Trans. Knowl. Data Eng. 32 (8) (2019) 1517–1528.
[25] Z. Zhang, W. Li, W. Ding, L. Zhang, Q. Lu, P. Hu, T. Gui, S. Lu, Stad-gan: unsupervised anomaly detection on multivariate time series with self-training generative adversarial networks, ACM Trans. Knowl. Discov. Data 17 (5) (2023) 1–18.
[26] M. Iman, H.R. Arabnia, K. Rasheed, A review of deep transfer learning and recent advancements, Technologies 11 (2) (2023) 40.
[27] H. Lu, Y. Zhao, Y. Song, Y. Yang, G. He, H. Yu, Y. Ren, A transfer learning-based intrusion detection system for zero-day attack in communication-based train control system, Clust. Comput. 27 (2024) 8477–8492.
[28] T. Upender, B. Lal, R. Nagaraju, Transfer learning method for handling the intrusion detection system with zero attacks using machine learning and deep learning, in: Proceedings of the 5th International Conference on Information Management & Machine Intelligence, 2023, pp. 1–11.
[29] G. Michau, O. Fink, Unsupervised transfer learning for anomaly detection: application to complementary operating condition transfer, Knowl.-Based Syst. 216 (2021) 106816.
[30] K.M. Ali Alheeti, K. McDonald-Maier, Intelligent intrusion detection in external communication systems for autonomous vehicles, Syst. Sci. Control Eng. 6 (1) (2018) 48–56.
[31] X. Li, Z. Hu, M. Xu, Y. Wang, J. Ma, Transfer learning based intrusion detection scheme for internet of vehicles, Inf. Sci. 547 (2021) 119–135.
[32] A. Boualouache, Y. Ghamri-Doudane, et al., Zero-x: a blockchain-enabled open-set federated learning framework for zero-day attack detection in iov, IEEE Trans. Veh. Technol. 73 (2024) 1–16.
[33] A.A. Korba, A. Boualouache, B. Brik, R. Rahal, Y. Ghamri-Doudane, S.M. Senouci, Federated learning for zero-day attack detection in 5g and beyond v2x networks, in: ICC 2023-IEEE International Conference on Communications, IEEE, 2023, pp. 1137–1142.
[34] X. He, Q. Chen, L. Tang, W. Wang, T. Liu, Cgan-based collaborative intrusion detection for uav networks: a blockchain-empowered distributed federated learning approach, IEEE Internet Things J. 10 (1) (2022) 120–132.
[35] R. Ahmad, I. Alsmadi, W. Alhamdani, L. Tawalbeh, Zero-day attack detection: a systematic literature review, Artif. Intell. Rev. 56 (10) (2023) 10733–10811.
[36] M. Sarhan, S. Layeghy, M. Gallagher, M. Portmann, From zero-shot machine learning to zero-day attack detection, Int. J. Inf. Secur. 22 (4) (2023) 947–959.
[37] T. Zhou, Q. Li, H. Lu, Q. Cheng, X. Zhang, Gan review: models and medical image fusion applications, Inf. Fusion 91 (2023) 134–148.
[38] I. Deshpande, Y.-T. Hu, R. Sun, A. Pyrros, N. Siddiqui, S. Koyejo, Z. Zhao, D. Forsyth, A.G. Schwing, Max-sliced Wasserstein distance and its use for gans, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 10648–10656.
[39] Q. Hoang, T.D. Nguyen, T. Le, D. Phung, Mgan: training generative adversarial nets with multiple generators, in: International Conference on Learning Representations, 2018, pp. 1–24.
[40] J. Kamel, M.R. Ansari, J. Petit, A. Kaiser, I.B. Jemaa, P. Urien, Simulation framework for misbehavior detection in vehicular networks, IEEE Trans. Veh. Technol. 69 (6) (2020) 6631–6643.
[41] J. Kamel, M. Wolf, R.W. Van Der Hei, A. Kaiser, P. Urien, F. Kargl, Veremi extension: a dataset for comparable evaluation of misbehavior detection in vanets, in: ICC 2020-2020 IEEE International Conference on Communications (ICC), IEEE, 2020, pp. 1–6.
[42] T. Shenkar, L. Wolf, Anomaly detection for tabular data with internal contrastive learning, in: International Conference on Learning Representations, 2022, pp. 1–26.
[43] B. Zong, Q. Song, M.R. Min, W. Cheng, C. Lumezanu, D. Cho, H. Chen, Deep autoencoding Gaussian mixture model for unsupervised anomaly detection, in: International Conference on Learning Representations, 2018, pp. 1–19.
[44] L. Ruff, R.A. Vandermeulen, N. Görnitz, A. Binder, E. Müller, K.-R. Müller, M. Kloft, Deep Semi-Supervised Anomaly Detection, 2019, pp. 1–23, arXiv preprint arXiv:1906.02694.
[45] G. Pang, C. Shen, H. Jin, A. van den Hengel, Deep weakly-supervised anomaly detection, in: Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2023, pp. 1795–1807.
[46] H. Xu, Y. Wang, J. Wei, S. Jian, Y. Li, N. Liu, Fascinating supervisory signals and where to find them: deep anomaly detection with scale learning, in: International Conference on Machine Learning, PMLR, 2023, pp. 38655–38673.

[47] P.C. Ngo, A.A. Winarto, C.K.L. Kou, S. Park, F. Akram, H.K. Lee, Fence gan: towards better anomaly detection, in: 2019 IEEE 31st International Conference on Tools with Artificial Intelligence (ICTAI), IEEE, 2019, pp. 141–148.

[48] N. Moustafa, J. Slay, Unsw-nb15: a comprehensive data set for network intrusion detection systems (unsw-nb15 network data set), in: 2015 Military Communications and Information Systems Conference (MilCIS), IEEE, 2015, pp. 1–6.

[49] Y. Yin, J. Jang-Jaccard, W. Xu, A. Singh, J. Zhu, F. Sabrina, J. Kwak, Igrf-rfe: a hybrid feature selection method for mlp-based network intrusion detection on unsw-nb15 dataset, J. Big Data 10 (1) (2023) 15.

[50] J. Sinha, M. Manollas, Efficient deep cnn-bilstm model for network intrusion detection, in: Proceedings of the 2020 3rd International Conference on Artificial Intelligence and Pattern Recognition, 2020, pp. 223–231.

[51] M.Y. Abualhoul, O. Shagdar, F. Nashashibi, Visible light inter-vehicle communication for platooning of autonomous vehicles, in: 2016 IEEE Intelligent Vehicles Symposium (IV), IEEE, 2016, pp. 508–513.

[52] A. Moubayed, A. Shami, P. Heidari, A. Larabi, R. Brunner, Edge-enabled v2x service placement for intelligent transportation systems, IEEE Trans. Mob. Comput. 20 (4) (2020) 1380–1392.